

Wazuh FIM Tutorial: File-Integrity-Monitoring

1- Definition

File Integrity Monitoring (FIM) is an essential cybersecurity technology and process that monitors, detects, and alerts changes made to critical operating system files, application binaries, configuration files, and registry entries.

File Integrity Monitoring (FIM) helps us detect unauthorized changes, malware activity, or policy violations across critical files and directories.

2- Configuration of FIM on windows and Linux

Having already deployed our Windows agents and Linux agents, let's configure FIM for them

A- Configuration on windows

1. **Navigate to Groups:** Open the main menu, go to **Agents Management**, and click **Groups**.
2. **Select Windows Group:** Open your Windows group (or whichever group contains your target machines).
3. **Edit Configuration:** Click on **Files** and edit the agent configuration file (agent.conf).
4. **Add FIM Settings:** Below our existing Windows Defender log collection block, paste this FIM configuration block:

```
<!-- FIM Configuration For Windows Agents -->
<syscheck>
  <scan_on_start>yes</scan_on_start>
  <interval>60</interval>
  <disabled>no</disabled>
  <report_changes>yes</report_changes>
  <directories check_all="yes" realtime="yes">C:\Users\timtc\Downloads\Timsand</directories>
</syscheck>
```

Key Settings Breakdown:

- **scan_on_start:** Triggers a full scan whenever the system boots up.
- **interval:** Defines how often periodic checks run (e.g., every 60 seconds).
- **disabled="no":** Enables FIM and its **real-time detection** feature to report file changes instantly.

Using Wildcards for Directory Paths

Next, specify the directory you want to monitor. For instance, the **Downloads** folder is a common target since malware often lands there first.

Because usernames vary across machines (e.g., Tims, Sandrine, or 100+ distinct user profiles), avoid hardcoding a single static path. Instead, use wildcard variables so the rule dynamically covers the Downloads folder across **all** connected Windows machines.

Once updated, save the configuration file. C:\Users\timtc\Downloads

B- Configuring the Linux Agent

Now let's set up our Linux machine:

1. Go to **Files** under your Linux agent group and open agent.conf.
2. Paste the Linux FIM configuration block

```
<!-- FIM Configuration For Linux Agents -->
<syscheck>
  <scan_on_start>yes</scan_on_start>
  <interval>60</interval>
  <disabled>no</disabled>
  <!-- Real-time monitoring and change reporting applied as attributes -->
  <directories check_all="yes" realtime="yes" report_changes="yes">/home/tim/Desktop/secret</directories>
</syscheck>
```

3. —in this example, we'll monitor a secret folder located on the user's desktop.
4. Click **Save**.

Finally, restart the Wazuh manager to apply the updated configurations across all agents.

```
sudo systemctl restart wazuh-manager
```

1 Definition: What is FIM?

File integrity monitoring (FIM) is an essential cybersecurity technology and process that monitors, detects, and alerts changes made to critical operating system files, application binaries, configuration files, and registry entries.



FIM helps us detect unauthorized changes, malware activity, or policy violations across critical files and directories.



Detect Unauthorized Changes



Identify Malware Activity



Ensure Compliance



Improve Security Visibility

"Monitor today. A safer tomorrow."

2 Prerequisites

Before configuring FIM, make sure you have:

- Wazuh server/manager already installed and running.
- Windows agents deployed and connected.
- Linux agents deployed and connected.
- Basic understanding of agent groups and configuration files (agent.conf).



Wazuh Manager



Windows Agents



Linux Agents

Environment ready! Let's configure FIM.

3 How FIM Works



Common Use Cases

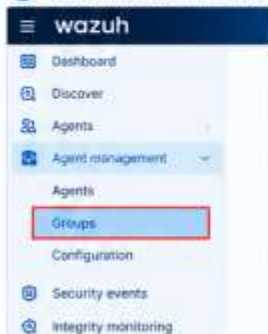
- Detecting malicious file modifications
- Monitoring configuration changes
- Tracking new executables (potential malware)
- Detecting deletion of critical system files
- Meeting compliance requirements (PCI-DSS, ISO 27001, etc.)

"File integrity monitoring: small changes, big impact on security."

Configure FIM on Windows Agents

4.1 Navigate to Groups

- Open the Wazuh dashboard.
- Click the menu (≡) on the top-left.
- Go to **Agents management** → **Groups**.



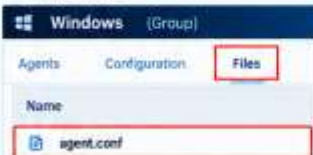
4.2 Select Windows Group

- Click on your Windows group (or the group containing your agents).

Name	Description	Agents
default	Default group	0
Windows	Windows machines	5
Linux	Linux machines	3

4.3 Edit Configuration

- Open the Windows group.
- Click on the **Files** tab and edit agent.conf.



4.4 Add FIM Configuration (Windows)

- Below your existing Windows Defender log collection block, paste the following FIM configuration:

```

<!-- FIM Configuration For Windows Agents -->
<syscheck>
  <scan_on_start>yes</scan_on_start>
  <interval>60</interval>
  <disabled>no</disabled>
  <report_changes>yes</report_changes>
  <directories check_all="yes" realtime="yes">
    C:\Users\timc\Downloads\Fimsand
  </directories>
</syscheck>
  
```

Key Settings Breakdown:

- scan_on_start**: Triggers a full scan whenever the system boots up.
- interval**: Defines how often periodic checks run (e.g., every 60 seconds).
- disabled="no"**: Enables FIM and its real-time detection feature.
- report_changes**: Reports file changes instantly.
- directories**: Specify the folder to monitor (use wildcards for all users).

Using Wildcards for Directory Paths

Because usernames vary across machines (e.g., Tims, Sandrine, or 100+ users), use a wildcard.

C:\Users*\Downloads

This will monitor the Downloads folder for all user profiles.

Save the configuration file.

5 Configure FIM on Linux Agents

5.1 Open Linux Group

- Go to **Agents management** → **Groups**.
- Click on your Linux group.

Name	Description	Agents
default	Default group	0
Windows	Windows machines	5
Linux	Linux machines	3

5.2 Add FIM Configuration (Linux)

- Go to the **Files** tab and open agent.conf.
- Paste the following configuration block:

```

<!-- FIM Configuration For Linux Agents -->
<syscheck>
  <scan_on_start>yes</scan_on_start>
  <interval>60</interval>
  <disabled>no</disabled>
  <!-- Real-time monitoring and change reporting -->
  <directories check_all="yes" realtime="yes">
    /home/tim/Desktop/secret/
  </directories>
  <report_changes>yes</report_changes>
</syscheck>
  
```

- In this example, we'll monitor a secret folder on the user's desktop.
- Click Save.

Configuration added for Linux agents.

6 Restart Wazuh Manager

- After updating configurations, restart the Wazuh manager to apply changes across all agents.

```
sudo systemctl restart wazuh-manager
```

Wazuh manager restarted successfully!

7 Verify & Test

- Perform a test (e.g., create, modify or delete a file in the monitored folder).
- Go to Wazuh dashboard → **Security events** to check for FIM alerts.



You should see the file change event in the dashboard.



Track Files
Detect Threats



Strengthen Security
With Open Source



Build Skills
Share Knowledge



Protect
What Matters



Author: **Tims TCHOUAMOU**
Cybersecurity Enthusiast | SOC Content Creator
Learn • Practice • Secure • Share

"Knowledge
Builds a Safer
Tomorrow"